

Data integrity and the transparency of processes: Each party to a Big Data analytics project must be aware of and abide by their responsibilities regarding the provision of source data and the obligation to establish and maintain adequate controls over the use of personal or other sensitive data.

Key challenges of IoT analytics applications

Heterogeneity: IoT data streams tend to be multi-modal and heterogeneous in their formats, semantics, and velocities. To access IoT data sources, database drivers and adaptors need to be deployed in Big Data applications. These applications incorporate features like interoperability, transaction management, and common data formats to address the heterogeneity.

Data quality: Different manufacturers provide various sensor categories and types. Substandard and unreliable sensors generate noisy and incomplete IoT data streams. Usage of these data streams leads to uncertainty and inaccuracy in IoT analytics. To improve the data quality and eliminate the noise, statistical models and probability based applications are developed to feed into IoT data analytics applications to generate an accurate output.

Real-time data sets: IoT devices provide high-speed and real-time data. In the Big Data ecosystem, data streaming platforms consume real-time data sets to generate valuable insights.

Location and time: IoT data comes with temporal and spatial metadata information, directly associated with its business value in each application context. These spatial and temporal dimensions can serve as a basis for dynamically selecting and filtering streams towards analytics applications for specific timelines and locations.

Security and privacy: IoT applications collect and process personal data and device data. IoT analytics helps in exhibiting strong IoT

security requirements and managing privacy sensitivities. The data generated by IoT data analytics needs to be encrypted, secured, and stored.

Data processing: IoT data analytics systems need to follow the steps listed below for data processing:

- Collection of data aggregated from a variety of sources, in a range of formats, and at multiple frequencies
- Understanding the data and categorising it
- Preparation of data
- Processing data from different sources
- Storing the information in time series for analysis
- Analysing the results
- Building applications to fulfil business processes

IoT analytics architecture

The requirements for an IoT analytics system with respect to its architecture are as follows.

- **Context:** Capture the context at all times, 24 hours a day, 365 days a year.
- **Standards:** Leverage standard protocols of communication between IoT devices and enterprise systems.
- **Scalability:** Always respond even during increased load; increase capacity proportionally as resources are added.
- **Data management:** Efficiently manage enormous volumes of data.
- **Connectivity:** Provide high network connectivity for large data payloads and continuous streaming.
- **Security:** Move and encrypt information securely even as IoT introduces new risks and vulnerabilities.
- **Interoperability:** Network all systems together and ensure interoperability of all data.
- **Device management:** An ability to connect to devices remotely at scale to manage them such as updating security credentials and firmware updates.

- **AI-driven analytics:** Enrich and explore prescriptive and predictive analytics data to deliver actionable insights.

Figure 1 shows the logical architecture of the IoT analytics system with key components and layers. While there exist many standard logical architectures for IoT analytics, this one is based on my practical experience across various domains and technologies.

The various layers for an enterprise application are classified as:

- Data source layer
- Edge layer
- Analytics layer
- Application layer

Data source layer: This layer consists of IoT devices and IoT gateways. Sensors and actuators are essential components of IoT.

Sensors sense the environment to monitor and detect changes of a particular attribute and collect the data, such as a change in temperature, heat, or cold. The output of the sensor is an electrical signal.

Actuators are devices used to control the system. They receive electrical output from sensors and produce a physical change. For example, if the sensor detects a change in temperature in a room, the actuator can turn a device on or off to maintain and control it.

These sensors and actuators establish a link between the physical and digital world via an IoT gateway. This layer basically deals with the gathering of specific data from data sources. This data is then passed to the edge layer so that it can be securely transmitted to the analytics layer.

Edge layer: The edge layer is an entry point for all things or devices to capture and process the data. The layer consists of a data store, services to analyse the data streamed from the IoT devices and execute AI-driven analytics locally in real-time, security mechanisms (including data encryption), and control applications.